Robert M. Gower

Talks

This is a complete list of the talks I have given at conferences, workshops, seminars, and summer schools, newest first. It is generated from `talks.bib`. More information at gowerrobert.github.io.

Upcoming

2026-11-06 **The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm**. Computational Mathematics and Scientific Computing Seminar, Courant Institute, New York University, New York, USA. [event].

2026-10-13 MINDS Seminar, Mathematical Institute for Data Science, Johns Hopkins University, Baltimore, USA. [event].

Past Talks

2026-07-09 **Non-Euclidean Gradient Descent Operates at the Edge of Stability**. International Conference on Machine Learning (ICML 2026), Oral, Seoul, South Korea. [event].

2026-06-02 **An Exploration of Non-Euclidean Gradient Descent: Muon and its Many Variants**. SIAM Conference on Optimization (OP26), minisymposium "Learning & Optimization", SIAM, Edinburgh, UK. [event].

2026-04-28 **The Polar Express**. Seminários Centro Pi, Instituto de Matemática Pura e Aplicada (IMPA), Rio de Janeiro, Brazil. [event | video].

2026-04-24 **The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm**. International Conference on Learning Representations (ICLR 2026), Oral, Rio de Janeiro, Brazil. Outstanding Paper Honorable Mention. [event | video].

2026-03-14 **The History of Mathematical Optimization in Blood and Booze**. Night in the Library, Brooklyn Public Library, Brooklyn, New York, USA. [event].

2026-02-06 **The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm**. Workshop on Randomized Numerical Linear Algebra, ICERM, Brown University, Providence, USA. [event | video].

2025-12-04 **In Search of Adam's Secret Sauce**. Conference on Neural Information Processing Systems (NeurIPS 2025), Oral, San Diego, USA. [event | video].

2022-10-28 **Practical Tools for Machine Learning and Data Science: Stochastic Gradient Descent, Tricks, Learning Rates, and Momentum**. Flatiron Wide Algorithms and Mathematics (FWAM), Flatiron Institute, New York, USA. Video available on the event page. [event].

2022-10-05 **Cutting Some Slack for SGD with Adaptive Polyak Stepsizes**. Learning and Optimization in Luminy (LOL 2022), CIRM, Marseille, France. [event].

2022-06-06 **Optimization for ML: First-Order Methods; Optimization for ML: Stochastic Gradient Descent**. Machine Learning X Science Summer School, Flatiron Institute, New York, USA. Two lectures, June 6–7, 2022. [event].

2022-04-29 **Cutting Some Slack for SGD with Adaptive Polyak Stepsizes**. ORFE Seminar, Operations Research and Financial Engineering, Princeton University, Princeton, USA. [event].

2021-01-27 **New Viewpoints, Variants and Convergence Theory for Stochastic Polyak Stepsizes**. MTL MLOpt Seminar, Montreal Machine Learning and Optimization, Online. [event | video].

2020-04-28 **Optimization for Machine Learning**. Mini-lecture series, Cornell University, Online. Three 50-minute lectures, April 28 – May 5, 2020. [event].

2019 Journée Statistique et Apprentissage, Institut des Hautes Études Scientifiques (IHÉS), Bures-sur-Yvette, France.

2019 Parietal Team Seminar, Inria Saclay, Saclay, France.

2019-08-07 **Expected Smoothness is the Key to Understanding the Mini-Batch Complexity of Stochastic Gradient Methods**. International Conference on Continuous Optimization (ICCOPT 2019), invited session, Mathematical Optimization Society, Berlin, Germany. [event | slides].

2019-06 **SGD: General Analysis and Improved Rates**. International Conference on Machine Learning (ICML 2019), Oral, Long Beach, USA. [event | video].

2018 Séminaire Parisien d'Optimisation, Institut Henri Poincaré, Paris, France.

2018-07 International Symposium on Mathematical Programming (ISMP 2018), invited session on Fast Converging Stochastic Optimization Algorithms, Mathematical Optimization Society, Bordeaux, France.

2018-06-26 Machine Learning and Optimization Group Seminar, EPFL, Lausanne, Switzerland.

2018-04 **Tracking the Gradients Using the Hessian: A New Look at Variance Reducing Stochastic Methods**. International Conference on Artificial Intelligence and Statistics (AISTATS 2018), semi-plenary oral, Lanzarote, Spain. [slides].

2017 PGMO Days, EDF Lab Paris-Saclay, Paris, France. [event].

2017 Optimization 2017, Faculdade de Ciências, Universidade de Lisboa, Lisbon, Portugal. [event].

2017 France/Japan Machine Learning Workshop, École Normale Supérieure, Paris, France. [event].

2017 Probabilistic Numerics Seminar, Max Planck Institute for Intelligent Systems, Tübingen, Germany.

2017 Operations Research Seminar, CORE, Université catholique de Louvain, Louvain-la-Neuve, Belgium.

2017-09-21 Optimization, Machine Learning, and Pluri-Disciplinarity Workshop, Inria Grenoble Rhône-Alpes, Grenoble, France. [event].

2017-06 **Randomized Iterative Methods for Computing the Pseudoinverse**. 27th Biennial Numerical Analysis Conference, University of Strathclyde, Glasgow, UK. [event | slides].

2017-06 **Randomized Iterative Methods for Linear Systems**. 18th IMA Leslie Fox Prize Meeting in Numerical Analysis, Institute of Mathematics and its Applications, University of Strathclyde, Glasgow, UK. Leslie Fox Prize, 2nd place. [event | slides].

2016 Computational Mathematics and Applications Seminar, Rutherford Appleton Laboratory, Oxfordshire, UK.

2016 Machine Learning and Statistics Seminar, Télécom ParisTech, Paris, France.

2016 5th IMA Conference on Numerical Linear Algebra and Optimization, University of Birmingham, Birmingham, UK.

2016 **Randomized Iterative Methods for Linear Systems**. Cambridge Image Analysis Group Seminar, University of Cambridge, Cambridge, UK.

2016 **Randomized Iterative Methods for Linear Systems and Inverting Matrices**. SIERRA Team Seminar, Inria, Paris, France.

2016-08 International Conference on Continuous Optimization (ICCOPT 2016), Mathematical Optimization Society, Tokyo, Japan.

2016-06 **Stochastic Block BFGS: Squeezing More Curvature out of Data**. International Conference on Machine Learning (ICML 2016), New York, USA. [slides].

2015 **Distributed Randomized Iterative Methods for Linear Systems**. Distributed Machine Learning and Optimization, Alan Turing Institute Scoping Workshop, ICMS, Edinburgh, UK. [event].

2015 **Randomized Iterative Methods for Linear Systems and Inverting Matrices**. Optimization and Big Data 2015, University of Edinburgh, Edinburgh, UK.

2015-07 **Action Constrained Quasi-Newton Methods**. International Symposium on Mathematical Programming (ISMP 2015), Mathematical Optimization Society, Pittsburgh, USA. [slides].

2014 **Generalizations of the Quasi-Newton Methods Using an Image Constraint**. European Workshop on Advances in Continuous Optimization (EUROPT 2014), EUROPT, Perpignan, France. [event | slides].

2014 **Hunting Inverses of Matrix Fields**. Irish Applied Mathematics Research Students' Meeting, SIAM Student Chapter, NUI Galway, Galway, Ireland. Best Talk Prize. [event | slides].

2014 **The History of Optimization in Blood and Booze**. Postgraduate Mathematics Colloquium, University of Edinburgh, Edinburgh, UK. [event].

2014 **Unconstrained Optimization Methods**. Research in the UK Afternoon, Cambridge University Mathematics Society, Cambridge, UK.

2013-07 **Third Order Methods and Third Order Derivatives**. International Conference on Continuous Optimization (ICCOPT 2013), Mathematical Optimization Society, Lisbon, Portugal. [event | slides].

2013-07 **Third Order Methods Using Slices of the Tensor and AD Developments**. EURO-INFORMS Joint International Meeting, EURO / INFORMS, Rome, Italy.

2011-07 **Automatic Differentiation of Hessian Matrices**. 28th Brazilian Colloquium on Mathematics, IMPA, Rio de Janeiro, Brazil. [slides].